```
define i8 @f(i8 %x) {
entry:
  %x = add i8 %x, 0
  ret i8 %x
}
```

$\sqsubseteq$

```
define i8 @f(i8 %x) {
entry:
  ret i8 %x
}
```

We do not model the **SSA** IR

$x + 0 \sqsubseteq x$

i.e., $x + 0$ *is refined by* $x$

# Verified Peephole Optimizer for Arithmetic Expressions in Lean 4

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# Peephole Optimization

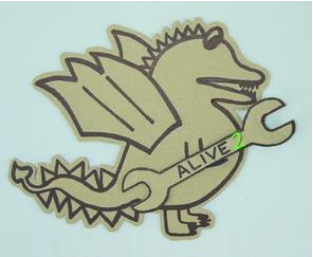
- Optimize code in a **small window** (peephole)

$$x + 0 \rightarrow x$$

```
define i8 @f(i8 %x) {
entry:
  %x = add i8 %x, 0
  ret i8 %x
}
```

$\sqsubseteq$

```
define i8 @f(i8 %x) {
entry:
  ret i8 %x
}
```




# Alive2

provides a **translation validation** tool to perform **refinement** check between two LLVM IR functions

based on **Z3** (an SMT Solver)

COMPILER EXPLORER

Add... More

LLVM IR source #1

LLVM IR

```
define i32 @src(i32 %x) {
  ret i32 %x
}

define i32 @tgt(i32 %x) {
  %res = add i32 %x, 1
  ret i32 %res
}
```

alive-tv (Editor #1, Compiler #2) LLVM IR

alive-tv

Compiler options...

```
----------------------------------------
define i32 @src(i32 %x) {
#0:
  ret i32 %x
}
=>
define i32 @tgt(i32 %x) {
#0:
  %res = add i32 %x, 1
  ret i32 %res
}
Transformation doesn't verify!

ERROR: Value mismatch

NOTE: The counterexample is unique.

Example:
i32 %x = #x00000003 (3)

Source:

Target:
i32 %res = #x00000004 (4)
Source value: #x00000003 (3)
Target value: #x00000004 (4)
```

https://alive2.llvm.org/ce/z/5RpnJi

# Undef

represents a value that can be any value when being evaluated

```
define i32 @src1() {
    ret i32 undef
}
```

```
define i32 @src2() {
    ret i32 undef
}
```

```
define i32 @src3() {
    ret i32 undef
}
```

```
define i32 @tgt1() {
    ret i32 0
}
```

```
define i32 @tgt2() {
    ret i32 1
}
```

```
define i32 @tgt3() {
    ret i32 -100
}
```

https://alive2.llvm.org/ce/z/jLhp4e

# Parameter Values Can Be Undef

```
define i8  @src(i8  %x) {
   %res = mul i8  %x , 2
   ret i8  %res
}
```

```
define i8  @tgt(i8  %x) {
   %res = add i8  %x , %x
   ret i8  %res
}
```

https://alive2.llvm.org/ce/z/_RdUm6

# Parameter Values Can Be Undef

different **undef**s can be evaluated to different values, even if they come from the same source (parameter)

```
define i8  @src(       ) {
   %res = mul i8  undef, 2
   ret i8  %res
}
```

```
define i8  @tgt(       ) {
   %res = add i8  undef, undef
   ret i8  %res
}
```

https://alive2.llvm.org/ce/z/S_zl9l

# Parameter Values Can Be Undef

different **undef**s can be evaluated to different values, even if they come from the same source (parameter)

```
define i8  @src(       ) {
   %res = mul i8  undef, 2
   ret i8  %res
}
```

$\not\sqsubseteq$

```
define i8  @tgt(       ) {
   %res = add i8  undef, undef
   ret i8  %res
}
```

`%res` can be any **even** i8 numbers

`%res` can be any i8 numbers

*Odd numbers are unexpected!*

https://alive2.llvm.org/ce/z/S_zl9l

# Alive2 Cannot Prove This!

Alive2 **timed out** while processing your query.

```
define i32 @src(        ) {
   %res = mul i32 undef, 2
   ret i32 %res
}
```

```
define i32 @tgt(        ) {
   %res = add i32 undef, undef
   ret i32 %res
}
```

https://alive2.llvm.org/ce/z/kLDrSq

# But We Can!

## even more general

Not

```
theorem undef_mul2_is_not_refined_by_undef_add_generic (w : Nat) (hpos : 0 < w) :
  ¬ ([llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = mul i$0 undef, 2
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)
  ⊑
  [llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = add i$0 undef, undef
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)) := by
```

# But We Can!

## even more general

```
theorem undef_mul2_is_not_refined_by_undef_add_generic (w : Nat) (hpos : 0 < w) :
  ¬ ([llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = mul i$0 undef, 2
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)
  ⊑
  [llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = add i$0 undef, undef
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)) := by
```

Integer with an **arbitrary width** represented by a **metavariable**

# But We Can!

## even more general

```
theorem undef_mul2_is_not_refined_by_undef_add_generic (w : Nat) (hpos : 0 < w) :
  ¬ ([llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = mul i$0 undef, 2
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)
  ⊑
  [llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = add i$0 undef, undef
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)) := by
```

Instantiate the **width metavariable** with **w**

# But We Can!

## even more general

```
theorem undef_mul2_is_not_refined_by_undef_add_generic (w : Nat) (hpos : 0 < w) :
  ¬ ([llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = mul i$0 undef, 2
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)
  ⊑   Is Refined By
  [llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = add i$0 undef, undef
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)) := by
```

# Refinement

x **is refined by** y if and only if ...

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔
```

# Refinement

... forall args ...

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
```

# Refinement

... such that everything is well-formed ...

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
```

# Refinement

... two conditions are satisfied

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
        ((
```

Condition 1: **Definedness**

```
        ) ∧ (
```

Condition 2: **Value Refinement**

```
        ))
```

# Refinement

Condition 1: Definedness
For all possible undef assignments, if x can be evaluated to a value, y should too

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
        ((
            (∀ (u : UndefChain), ∃ (ret_x : RegisterValue),
                run (eval x args) default ⟨u, 0⟩ = .ok ret_x) →
            (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
                run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
        ) ∧ (
```

Condition 2: **Value Refinement**

```
        ))
```

# Refinement

**Condition 2: Value Refinement**
For all values returned by y, it should refine the value returned by x

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
        ((
            (∀ (u : UndefChain), ∃ (ret_x : RegisterValue),
                run (eval x args) default ⟨u, 0⟩ = .ok ret_x) →
            (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
                run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
        ) ∧ (∀ (u : UndefChain) (ret : RegisterValue),
            run (eval y args) default ⟨u, 0⟩ = .ok ret →
                ReturnValuesRefined x args u ret)
        ))
```

# Refinement

This theorem is proved by **rfl** because this is just the definition of isRefinedBy

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
        ((
            (∀ (u : UndefChain), ∃ (ret_x : RegisterValue),
                run (eval x args) default ⟨u, 0⟩ = .ok ret_x) →
            (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
                run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
        ) ∧ (∀ (u : UndefChain) (ret : RegisterValue),
            run (eval y args) default ⟨u, 0⟩ = .ok ret →
                ReturnValuesRefined x args u ret)
        )) := by
        rfl
```

# Return Value Refinement

```
theorem definition_returnValuesRefined_iff x args u ret:
    ReturnValuesRefined x args u ret ↔
        ∃ u’, run (eval x args) default ⟨u', 0⟩ = .ok ret
        ∨
        ∃ ret_x,
            run (eval x args) default ⟨u', 0⟩ = .ok ret_x ∧
                RegisterValue.IsRefinedBy ret_x ret := by rfl
```

Or

Equivalent

Refined

```
def RegisterValue.IsRefinedBy : RegisterValue → RegisterValue → Prop
| .bv w_x .poison, .bv w_y _ => w_x = w_y
| .bv w_x (.value v_x), .bv w_y (.value v_y) => ∃ h : w_x = w_y, h ▸ v_x = v_y
| .void, .void => True
| _, _ => False
```

# How to Prove

```
theorem undef_mul2_is_not_refined_by_undef_add_generic (w : Nat) (hpos : 0 < w) :
  ¬ ([llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = mul i$0 undef, 2
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)
  ⊑
  [llvm-1-definition|
    define i$0 @f() {
      entry:
        %x = add i$0 undef, undef
        ret i$0 %x
    }
  ].instantiateWidths (singletonWidths w)) := by
```

```
intro h
```

```
have haddRun :
    run (eval ([llvm-1-definit
        define i$0 @f() {
        entry:
            %x = add i$0 undef
            ret i$0 %x
        }
    ].instantiateWidths (singl
        ({ supplyChain := [1,
    Except.ok (RegisterValue.b
        simp [simp_llvm]
        rw [<- BitVec.add_zero
        simp
```

**1 goal**

```
w : ℕ
hpos : 0 < w
h : AST.Definition.instantiateWidths (singletonWidths w)
    ([llvm-definition|define i$0 @f() {
    entry:
        %x = mul i$0 undef! 0, 2
        ret i$0 %x
    }]) ⊑
  AST.Definition.instantiateWidths (singletonWidths w)
    ([llvm-definition|define i$0 @f() {
    entry:
        %x = add i$0 undef! 0, undef! 1
        ret i$0 %x
    }])
⊢ False
```

```
have haddRun :
    run (eval ([llvm-1-definition|
        define i$0 @f() {
        entry:
            %x = add i$0 undef, undef
            ret i$0 %x
        }
    ].instantiateWidths (singletonWidths w)) []) default
        ({ supplyChain := [1, 0], supplyIndex := 0 }) =
    Except.ok (RegisterValue.bv w (.value (BitVec.ofNat w 1))) := by
        simp [simp_llvm]
        rw [<- BitVec.add_zero 1#w]
        simp
```

Evaluating RHS (**y**) with **1st undef = 1, 2nd undef = 0**

```
have haddRun :
    run (eval ([llvm-1-definition|
        define i$0 @f() {
        entry:
            %x = add i$0 undef, undef
            ret i$0 %x
        }
    ].instantiateWidths (singletonWidths w)) []) default
        ({ supplyChain := [1, 0], supplyIndex := 0 }) =
    Except.ok (RegisterValue.bv w (.value (BitVec.ofNat w 1))) := by
        simp [simp_llvm]
        rw [<- BitVec.add_zero 1#w]
        simp
```

Result is 1

```
have haddRun :
    run (eval ([llvm-1-definition|
        define i$0 @f() {
        entry:
            %x = add i$0 undef, undef
            ret i$0 %x
        }
    ].instantiateWidths (singletonWidths w)) []) default
        ({ supplyChain := [1, 0], supplyIndex := 0 }) =
    Except.ok (RegisterValue.bv w (.value (BitVec.ofNat w 1))) := by
        simp [simp_llvm]
        rw [<- BitVec.add_zero 1#w]
        simp
```

All **evalXXX** definitions are annotated with **[simp_llvm]**

```
    rw ⌊<- BitVec.add_zero 1#w⌋
    simp
```

The refinement

```
h : AST.Definition.instantiateWidths (singletonWidths w)
    ([llvm-definition|define i$0 @f() {
    entry:
        %x = mul i$0 undef! 0, 2
        ret i$0 %x
    }]) ⊑
  AST.Definition.instantiateWidths (singletonWidths w)
    ([llvm-definition|define i$0 @f() {
    entry:
        %x = add i$0 undef! 0, undef! 1
        ret i$0 %x
    }])
```

```
obtain ⟨u', hu'⟩ := (h []
    (by simp [simp_wellform])
    (by simp [simp_wellform])
    (by rfl)
    (by simp [Definition.ArgValuesWellFormed, simp_wellform])
).2 [1, 0] (RegisterValue.bv w (.value (BitVec.ofNat w 1))) haddRun
```

```
cases u' with
| nil =>
    have hnil :
        run (eval ([llvm-1-definition|
            define i$0 @f() {
```

Specialize h with empty args

and proofs for well-formedness

```
obtain ⟨u', hu'⟩ := (h []
    (by simp [simp_wellform])
    (by simp [simp_wellform])
    (by rfl)
    (by simp [Definition.ArgValuesWellFormed, simp_wellform])
).2 [1, 0] (RegisterValue.bv w ( value (BitVec.ofNat w 1))) haddRun
```

```
cases u' with
| nil =>
    have hnil :
        run (eval ([llvm-1-defi
            define i$0 @f() {
            entry:
                %x = mul i$0 und
                ret i$0 %x
```

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
        ((
            (∀ (u : UndefChain), ∃ (ret_x : RegisterValue),
                run (eval x args) default ⟨u, 0⟩ = .ok ret_x) →
            (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
                run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
        ) ∧ (∀ (u : UndefChain) (ret : RegisterValue),
            run (eval y args) default ⟨u, 0⟩ = .ok ret →
                ReturnValuesRefined x args u ret)
        )) := by
        rfl
```

```
obtain ⟨u', hu'⟩ := (h []
    (by simp [simp_wellform])
    (by simp [simp_wellform])
    (by rfl)
    (by simp [Definition.ArgValuesWellFormed, simp_wellform])
).2 [1, 0] (RegisterValue.bv w (.value (BitVec.ofNat w 1))) haddRun
```

Obtain the 2nd condition

```
cases u' with
| nil =>
    have hnil :
        run (eval ([llvm-1-defi
            define i$0 @f() {
            entry:
                %x = mul i$0 und
                ret i$0 %x
            }
```

```
theorem definition_isRefinedBy_iff (x y : @AST.Definition ϕ) :
    x ⊑ y ↔ (∀ (args : List RegisterValue),
        WellFormed x → WellFormed y →
        SignatureCompatible x y → ArgValuesWellFormed x args →
        ((
            (∀ (u : UndefChain), ∃ (ret_x : RegisterValue),
                run (eval x args) default ⟨u, 0⟩ = .ok ret_x) →
            (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
                run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
        ) ∧ (∀ (u : UndefChain) (ret : RegisterValue),
            run (eval y args) default ⟨u, 0⟩ = .ok ret →
                ReturnValuesRefined x args u ret)
        )) := by
        rfl
```

```
obtain ⟨u', hu'⟩ := (h []
    (by simp [simp_wellform])
    (by simp [simp_wellform])
    (by rfl)
    (by simp [Definition.ArgValuesWellFormed, simp_wellform])
).2 [1, 0] (RegisterValue.bv w (.value (BitVec.ofNat w 1))) haddRun
```

Specialize it with

- undef supply chain = [1, 0]
- ret = 1
- A prop stating that RHS (y) returns ret

```
| nil =>
    have hnil :
        run (eval ([llvm-1-defi
            define i$0 @f() {
            entry:
                %x = mul i$0 und
                ret i$0 %x
            }
```

```
em definition_isRefinedBy_iff (x y : @AST.Definition φ) :
 x ⊑ y ↔ (∀ (args : List RegisterValue),
      WellFormed x → WellFormed y →
      SignatureCompatible x y → ArgValuesWellFormed x args →
      ((
            (∀ (u : UndefChain), ∃ (ret_x : RegisterValue),
                run (eval x args) default ⟨u, 0⟩ = .ok ret_x) →
            (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
                run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
      ) ∧ (∀ (u : UndefChain) (ret : RegisterValue),
            run (eval y args) default ⟨u, 0⟩ = .ok ret →
                ReturnValuesRefined x args u ret)
      )) := by
      rfl
```

Now, we got

- u’, the supply chain for LHS (x)
- hu’, a prop stating that by evaluating LHS (x) under the undef supply u’, either
  - LHS (x) also returns ret, or
  - the return value (ret_x) of LHS (x) is refined by ret

```
obtain ⟨u', hu'⟩ := (h []
    (by simp [simp_wellform])
    (by simp [simp_wellform])
    (by rfl)
    (by simp [Def
).2 [1, 0] (Regis
```

```
theorem definition_returnValuesRefined_iff x args u ret:
    ReturnValuesRefined x args u ret ↔
        ∃ u’, run (eval x args) default ⟨u', 0⟩ = .ok ret
              ∨
              ∃ ret_x,
                  run (eval x args) default ⟨u', 0⟩ = .ok ret_x ∧
                      RegisterValue.IsRefinedBy ret_x ret
```

```
cases u' with
| nil =>
    have hnil :
        run (eval ([llvm-1-defi
            define i$0 @f() {
            entry:
                %x = mul i$0 und
```

```
        (∀ (u' : UndefChain), ∃ (ret_y : RegisterValue),
            run (eval y args) default ⟨u', 0⟩ = .ok ret_y)
    ) ∧ (∀ (u : UndefChain) (ret : RegisterValue),
        run (eval y args) default ⟨u, 0⟩ = .ok ret →
            ReturnValuesRefined x args u ret)
    )) := by
    rfl
```

Case analysis on the supply chain

```
cases u' with
| nil =>
    have hnil :
        run (eval ([llvm-1-definition|
            define i$0 @f() {
            entry:
                %x = mul i$0 undef, 2
                ret i$0 %x
            }
        ].instantiateWidths (singletonWidths w)) []) default
            ({ supplyChain := [], supplyIndex := 0 }) =
        Except.error "" := by
            simp
            rfl
    simp_all
```

```
cases u' with
| nil =>
    have hnil :
        run (eval ([llvm-1-definition|
            define i$0 @f() {
            entry:
                %x = mul i$0 undef, 2
                ret i$0 %x
            }
        ].instantiateWidths (singletonWidths w)) []) default
            ({ supplyChain := [], supplyIndex := 0 }) =
        Except.error "" := by
            simp
            rfl
    simp_all
```

When the supply chain is empty, LHS will throw an error, because there is an undef value in LHS expecting a supply.

So, the hypothesis is false
and can be easily completed by `simp_all`

```
| cons x xs =>
    have hmulRun :
        run (eval ([llvm-1-definition|
            define i$0 @f() {
            entry:
                %x = mul i$0 undef, 2
                ret i$0 %x
            }
        ].instantiateWidths (singletonWidths w)) []) default
            ({ supplyChain := x :: xs, supplyIndex := 0 }) =
        Except.ok (RegisterValue.bv w (.value ((BitVec.ofNat w x) * (2#w)))) := by
            simp [simp_llvm]
            rfl
```

When the supply chain contains at least one element x,

LHS returns $x * 2$

```
simp_all
```

After `simp_all`, hu' (the refinement) is simplified to $x * 2 = 1$

```
have hcontra := bitvec_mul_two_ne_one_generic hpos (BitVec.ofNat w x)
contradiction
```

which contradicts a lemma saying that $x * 2 \neq 1$

# Summary

https://github.com/xuhongxu96/lean-nano-llvm

**Achievements**

- an AST for a LLVM-like IR
- a DSL embedded in Lean
  - with unexpanders
- well-formedness checks
- big-step operational semantics
- a supply-chain model of undef
- a semantic refinement relation
- some example theorem
  - including width-generic examples

**Limitations for Future Work**

- No memory model
  - load and store
  - pointer reasoning
- No optimizer function
  - or relevant proofs
- No control flow
  - only support a single basic block
- Incomplete undef support
- Incomplete instruction support